

Student Name: _____

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **40**Instructor Name: **Mamani Velasco, Ruben****Question 1 of 40**

Find an equation for the line that passes through the points $(6, -4)$ and $(-2, 6)$.

Question 2 of 40

Solve for x .

$$x + \frac{15}{x} = -8$$

Question 3 of 40

A local hamburger shop sold a combined total of 767 hamburgers and cheeseburgers on Saturday. There were 67 more cheeseburgers sold than hamburgers. How many hamburgers were sold on Saturday?

Question 4 of 40

An item on sale costs 15% of the original price. The original price was \$61.

Use a calculator to find the sale price.

Question 5 of 40

The functions f and g are defined as follows.

$$f(x) = \frac{x + 7}{x^2 + 16x + 63}$$

$$g(x) = \frac{x^2}{x - 8}$$

For each function, find the domain.

Write each answer as an interval or union of intervals.

Domain of f : _____

Domain of g : _____

Question 6 of 40

For the real-valued functions $g(x) = x^2 + 1$ and $h(x) = \sqrt{x - 4}$, find the composition $g \circ h$. Also, specify its domain using interval notation.

$(g \circ h)(x) =$

Domain of $g \circ h$:

For the real-valued functions $g(x) = x^2 + 1$ and $h(x) = \sqrt{x - 4}$, find the composition $g \circ h$. Also, specify its domain using interval notation.

$g \circ h \ x =$

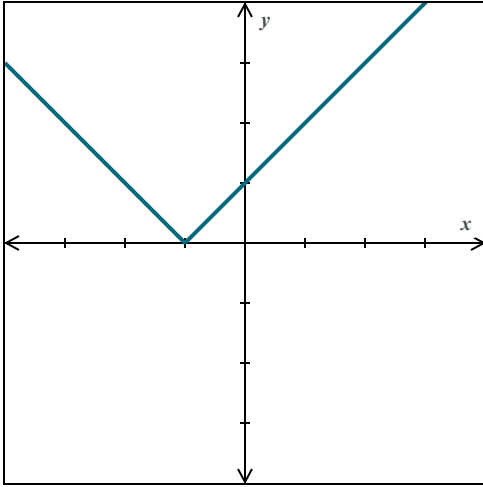
Domain of $g \circ h$:

Question 7 of 40

Four functions are given below. Either the function is defined explicitly, or the entire graph of the function is shown.

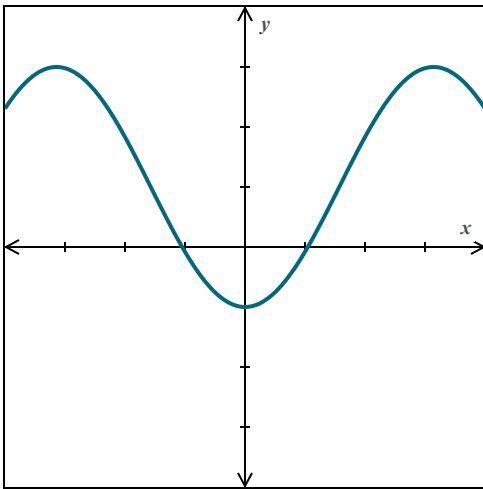
For each, decide whether it is an even function, an odd function, or neither.

The function r



- Even - Odd - Neither

The function s



- Even - Odd - Neither

$$g(x) = -6x^4 + 3x^2$$

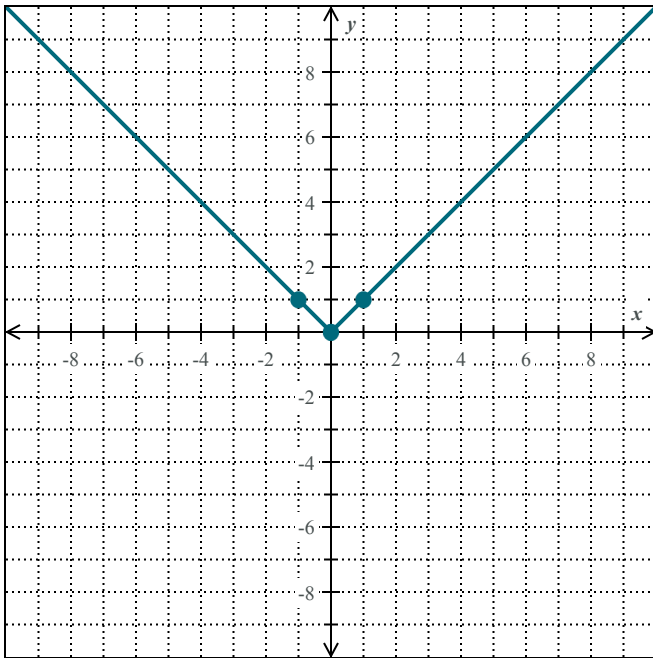
- Even - Odd - Neither

$$h(x) = -2x^5 + 5x^2$$

- Even - Odd - Neither

Question 8 of 40

Below is the graph of $y = |x|$. Translate it to make it the graph of $y = |x + 4| + 1$.



Question 9 of 40

Subtract.

$$(-3 + 5i) - (-4 + 4i)$$

Write your answer as a complex number in standard form.

Question 10 of 40

Multiply.

$$(3 - 6i)(1 - i)$$

Write your answer as a complex number in standard form.

Question 11 of 40

Solve the quadratic equation by completing the square.

$$x^2 + 4x - 6 = 0$$

(a) Choose the appropriate form and fill in the blanks with the correct numbers.

☐ $(x + \underline{\hspace{1cm}})^2 = \underline{\hspace{1cm}}$

☐ $(x - \underline{\hspace{1cm}})^2 = \underline{\hspace{1cm}}$

(b) Solve the equation. Simplify your answer as much as possible. If there is more than one solution, separate them with commas.

$x = \underline{\hspace{3cm}}$

Question 12 of 40

The length of a rectangle is 1 yd more than double the width, and the area of the rectangle is 45 yd^2 .

Find the dimensions of the rectangle.

Length: yd

Width: yd

Question 13 of 40

Solve for y .

$$3|2y + 8| - 37 = -7$$

Question 14 of 40

Solve for y , where y is a real number.

$$\sqrt{3y - 14} + 4 = 2$$

Question 15 of 40

Suppose that the function f is defined on the interval $(-2, 2]$ as follows.

$$f(x) = \begin{cases} -1 & \text{if } -2 < x \leq -1 \\ 0 & \text{if } -1 < x \leq 0 \\ 1 & \text{if } 0 < x \leq 1 \\ 2 & \text{if } 1 < x \leq 2 \end{cases}$$

Find $f(-1)$, $f(-0.5)$, and $f(1)$.

$f(-1) =$	
$f(-0.5) =$	
$f(1) =$	

Question 16 of 40

Divide.

$$(9x^3 - 6x^2 + 9x + 10) \div (3x^2 + x)$$

Your answer should give the quotient and the remainder.

Quotient:

Remainder:

Question 17 of 40

Use synthetic division to find the quotient and remainder when $5x^3 + 21x^2 - 16$ is divided by $x + 4$ by completing the parts below.

(a) Complete this synthetic division table.

-4)	5	21	0	-16

(b) Write your answer in the following form: Quotient + $\frac{\text{Remainder}}{x + 4}$.

$5x^3 + 21x^2 - 16$	=		+	
$x + 4$				$x + 4$

Question 18 of 40

Consider the following polynomial function.

$$f(x) = (x - 1)(x + 1)(x - 2)^2$$

Answer the questions regarding the graph of f .
Then, use this information to graph the function.

(a) Choose the end behavior of the graph of f .

(Blank 1,)

Blank 1, Options

- Falls to the left and rises to the right
- Rises to the left and falls to the right
- Rises to the left and rises to the right
- Falls to the left and falls to the right

(b) List each real zero of f according to the behavior of the graph at the x -axis near that zero. If there is more than one answer, separate them with commas. If there is no answer, write "None".

Zero(s) where the graph crosses the x -axis:

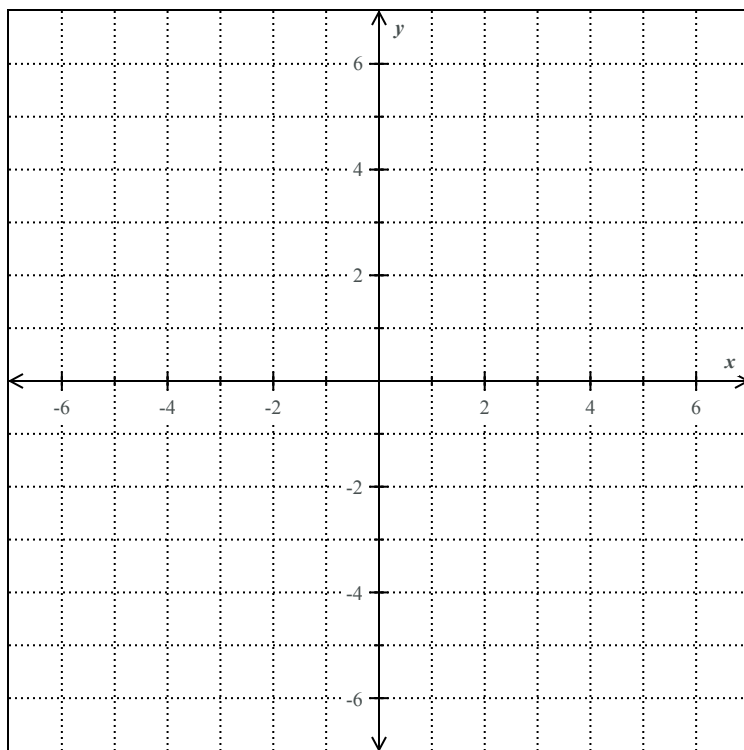
Zero(s) where the graph touches, but does not cross the x -axis:

(c) Find the y -intercept of the graph of f .

y -intercept:

(d) Graph $f(x) = (x - 1)(x + 1)(x - 2)^2$ by doing the following.

- Plot all points where the graph of f intersects the x -axis or y -axis.
- For each point on the x -axis, select the correct behavior.
- Connect the points following the behaviors.



Question 19 of 40

For the polynomial below, -2 is a zero.

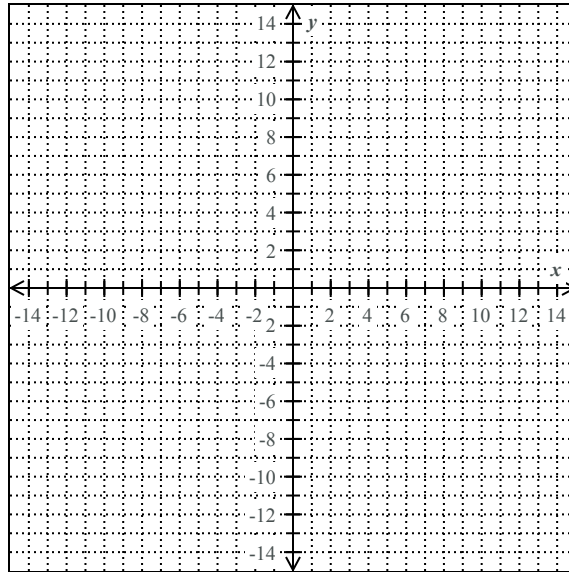
$$h(x) = x^3 - 2x^2 - 5x + 6$$

Express $h(x)$ as a product of linear factors.

Question 20 of 40

Graph the rational function.

$$f(x) = \frac{-12}{x^2 + 8x + 12}$$

**Question 21 of 40**

Solve the following inequality.

$$\frac{x-4}{x+7} \leq 0$$

Write your answer using interval notation.

Question 22 of 40

Solve the following inequality.

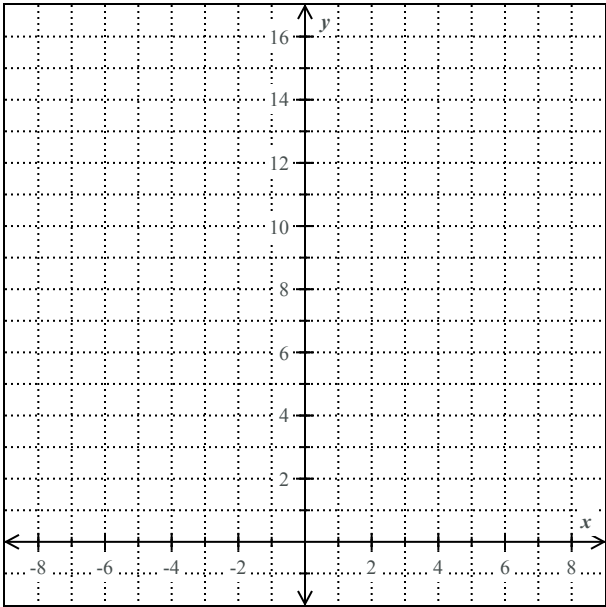
$$(x-1)(x+4)(x-5) \leq 0$$

Write your answer as an interval or union of intervals.

Question 23 of 40

Graph the exponential function.

$$f(x)=\left(\frac{4}{3}\right)^x$$



Question 24 of 40

The radioactive substance uranium-240 has a half-life of 14 hours. The amount $A(t)$ of a sample of uranium-240 remaining (in grams) after t hours is given by the following exponential function.

$$A(t)=2,400\left(\frac{1}{2}\right)^{\frac{t}{14}}$$

Find the initial amount in the sample and the amount remaining after 40 hours.
Round your answers to the nearest gram as necessary.

Initial amount: grams

Amount after 40 hours: grams

Question 25 of 40

Rewrite each equation as requested.

(a) $\square^{\square} = \square$

(b) $\log_{\square} \square = \square$

(a) Rewrite as an exponential equation.

$$\log_7 \frac{1}{7} = -1$$

$$\square^{\square} = \square$$

(b) Rewrite as a logarithmic equation.

$$2^5 = 32$$

$$\log_{\square} \square = \square$$

Question 26 of 40

Evaluate each expression.

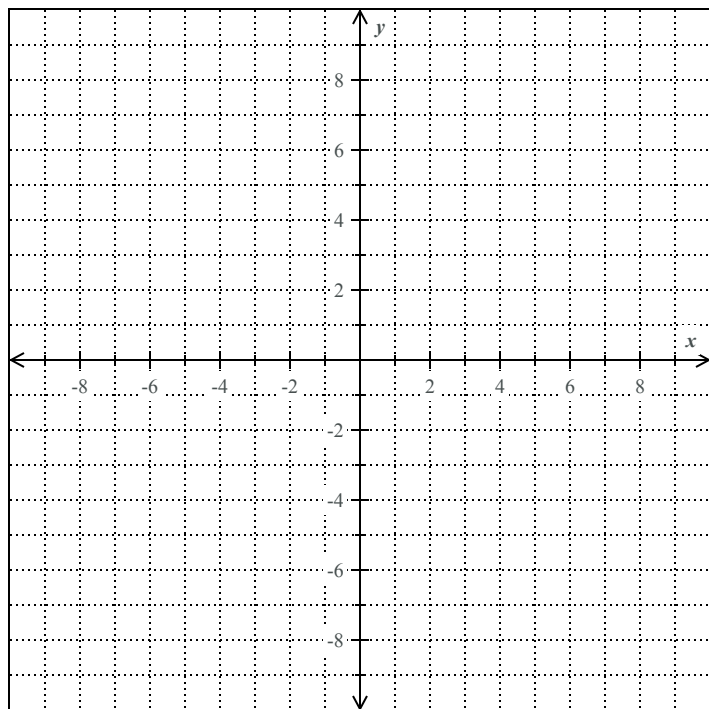
(a) $\log_9 1 = \square$

(b) $\log_8 \frac{1}{64} = \square$

Question 27 of 40

Graph the logarithmic function.

$$g(x) = 1 + \log_2 x$$

**Question 28 of 40**

Fill in the missing values to make the equations true.

$$\log_4 7 - \log_4 11 = \log_4 \boxed{}$$

$$\log_5 \boxed{} + \log_5 9 = \log_5 99$$

$$\log_2 \frac{1}{25} = \boxed{} \log_2 5$$

Question 29 of 40

Use the properties of logarithms to expand the following expression.

$$\log \left(x^5 \sqrt[3]{y^2 z} \right)$$

Each logarithm should involve only one variable and should not have any radicals or exponents.

You may assume that all variables are positive.

$$\log \left(x^5 \sqrt[3]{y^2 z} \right) = \boxed{}$$

Question 30 of 40

Write the expression as a single logarithm.

$$7 \log_5 (2z + 1) + \frac{1}{4} \log_5 (z + 6)$$

$$\log \underline{\hspace{2cm}} \left(\underline{\hspace{2cm}} \right)$$

Question 31 of 40

The number of bacteria in a certain sample increases according to the following function, where y_0 is the initial number present, and y is the number present at time t (in hours).

$$y = y_0 e^{0.059t}$$

How many hours does it take for the size of the sample to double? Do not round any intermediate computations, and round your answer to the nearest tenth.

Question 32 of 40

Solve for x .

$$5 + \ln (x + 4) = 8$$

Do not round any intermediate computations, and round your answer to the nearest hundredth.

Question 33 of 40

Solve for x .

$$\log_2(x + 7) = 3 - \log_2(x + 5)$$

$x =$

Question 34 of 40

Use substitution to solve the system.

$$\begin{cases} 4x - 5y = 37 \\ y = 3x - 14 \end{cases}$$

$x =$	
$y =$	

Question 35 of 40

Kira the trainer has two solo workout plans that she offers her clients: Plan A and Plan B. Each client does either one or the other (not both). On Friday there were 3 clients who did Plan A and 5 who did Plan B. On Saturday there were 6 clients who did Plan A and 2 who did Plan B. Kira trained her Friday clients for a total of 10 hours and her Saturday clients for a total of 10 hours. How long does each of the workout plans last?

Question 36 of 40

Consider the following system of linear equations.

$$2x - 12y = 16$$

$$4x - 19y = 22$$

Solve the system by completing the steps below to produce a reduced row-echelon form. R_1 and R_2 denote the first and second rows, respectively. The arrow notation ($\rightarrow$) means the expression/matrix on the left becomes the expression/matrix on the right once the row operations are performed.

(a) For each step below, enter the coefficient for the row operation. The first matrix in Step 1 is the augmented matrix for the given system of equations.

Step 1:

$$\left[\begin{array}{cc|c} 2 & -12 & 16 \\ 4 & -19 & 22 \end{array} \right] \xrightarrow{\text{---} \cdot R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & -6 & 8 \\ 4 & -19 & 22 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{cc|c} 1 & -6 & 8 \\ 4 & -19 & 22 \end{array} \right] \xrightarrow{\text{---} \cdot R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 5 & -10 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 5 & -10 \end{array} \right] \xrightarrow{\text{---} \cdot R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 1 & -2 \end{array} \right]$$

Step 4: Enter the coefficient for the row operations and the missing entries in the resulting matrix.

$$\left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 1 & -2 \end{array} \right] \xrightarrow{\text{---} \cdot R_2 + R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & 0 & \text{---} \\ 0 & 1 & \text{---} \end{array} \right]$$

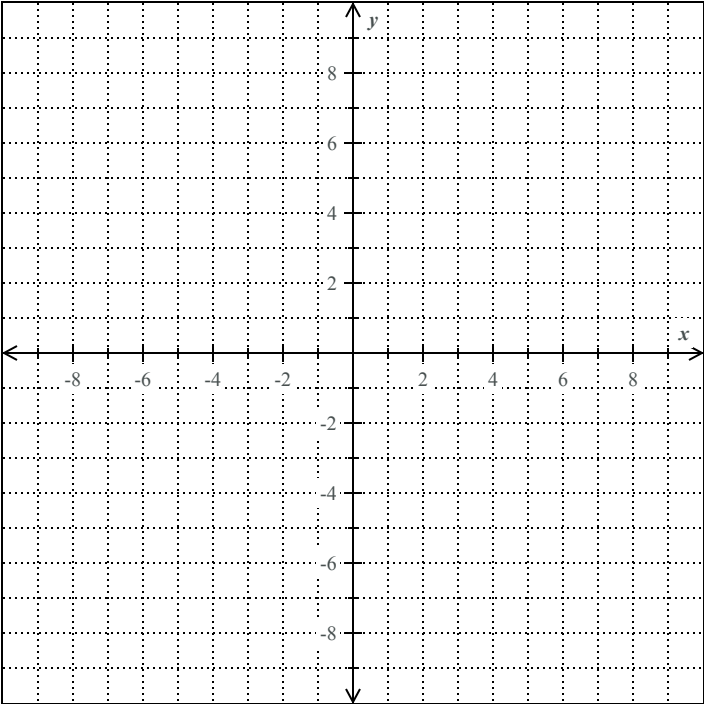
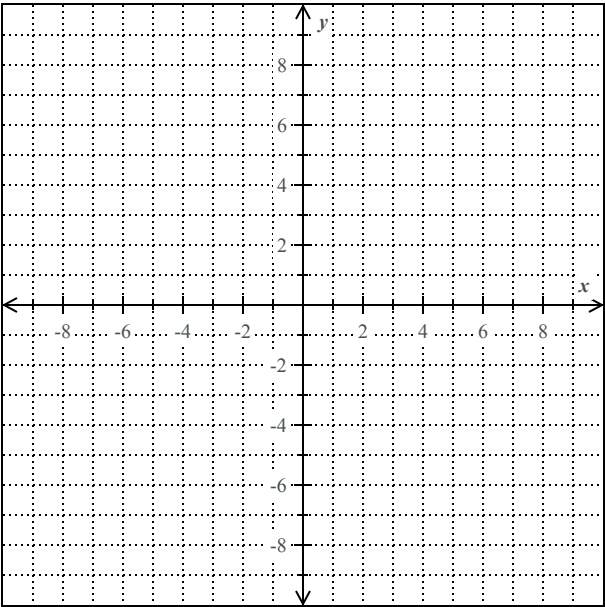
(b) Give the solution.

$x = \text{---}$ $y = \text{---}$

Question 37 of 40

Graph the ellipse.

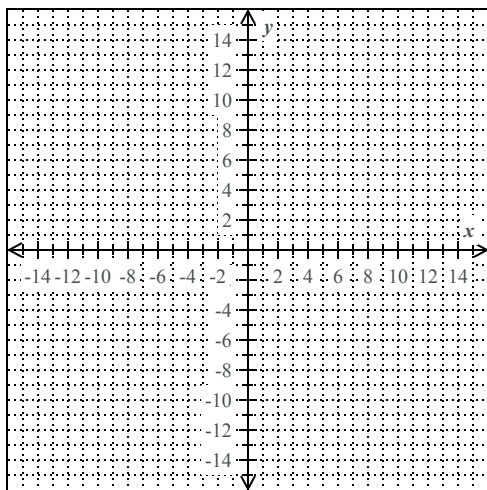
$$\frac{(x - 3)^2}{25} + \frac{(y + 4)^2}{4} = 1$$



Question 38 of 40

Graph the hyperbola.

$$\frac{(x+2)^2}{25} - \frac{(y-3)^2}{4} = 1$$

**Question 39 of 40**

Find the first four terms of the sequence given by the following.

$$a_n = \frac{3^n + 5}{3^n}, n = 1, 2, 3, \dots$$

Question 40 of 40

Compute each sum below. If applicable, write your answer as a fraction.

$$\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \dots + \left(\frac{1}{3}\right)^7$$

$$\sum_{j=1}^8 2(-4)^j$$

Compute each sum below. If applicable, write your answer as a fraction.

$$\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \dots + \left(\frac{1}{3}\right)^7 = \underline{\hspace{2cm}}$$

$$\sum_{j=1}^8 2(-4)^j = \underline{\hspace{2cm}}$$

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Number of Questions: **40**

Question 1 of 40

$$y = -\frac{5}{4}x + \frac{7}{2}$$

Note: Other correct answers include $y + 4 = -\frac{5}{4}(x - 6)$ and $5x + 4y = 14$.

Question 2 of 40

$$x = -5, -3$$

Question 3 of 40

350 hamburgers

Question 4 of 40

\$9.15

Question 5 of 40

Domain of f : $(-\infty, -9) \cup (-9, -7) \cup (-7, \infty)$

Domain of g : $(-\infty, 8) \cup (8, \infty)$

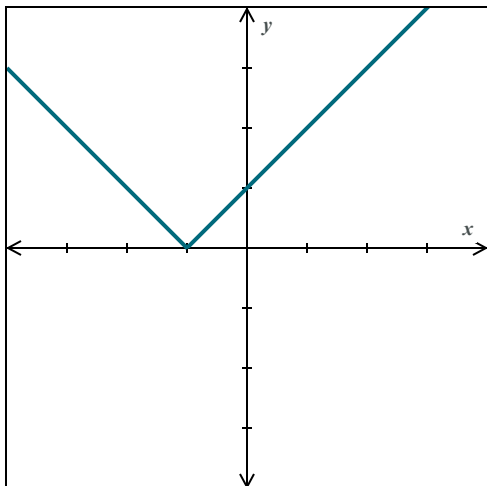
Question 6 of 40

$$(g \circ h)(x) = x - 3$$

Domain of $g \circ h$: $[4, \infty)$

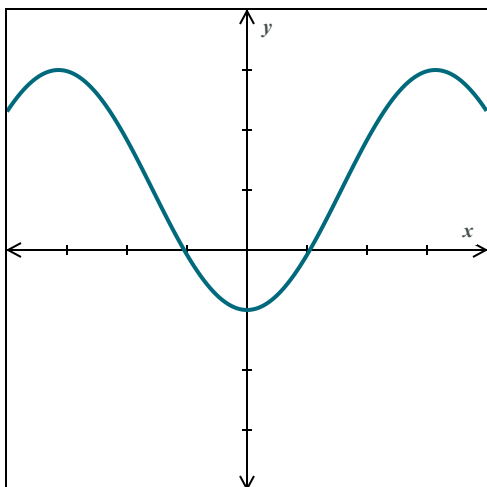
Question 7 of 40

The function r



- Neither

The function s



- Even

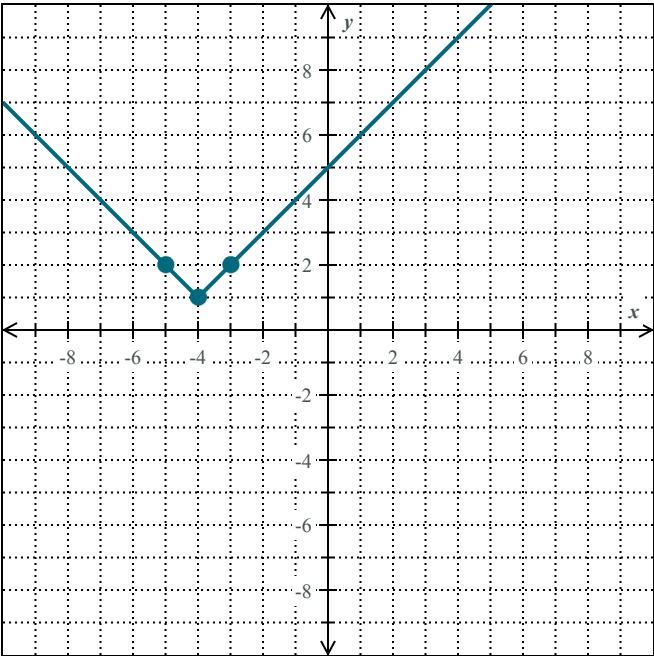
$$g(x) = -6x^4 + 3x^2$$

- Even

$$h(x) = -2x^5 + 5x^2$$

- Neither

Question 8 of 40



Question 9 of 40

$1 + i$

Question 10 of 40

$-3 - 9i$

Question 11 of 40

(a) ☒ $(x + 2)^2 = 10$

☐ $(x - \boxed{})^2 = \boxed{}$

(b) $x = -2 + \sqrt{10}, -2 - \sqrt{10}$

Question 12 of 40

Length: 10 yd

Width: 4.5 yd

Question 13 of 40

$$y = 1, -9$$

Question 14 of 40

No solution

Question 15 of 40

$$f(-1) = -1$$

$$f(-0.5) = 0$$

$$f(1) = 1$$

Question 16 of 40

Quotient: $3x - 3$

Remainder: $12x + 10$

Question 17 of 40

(a) Complete this synthetic division table.

$$\begin{array}{r|rrrr} -4 & 5 & 21 & 0 & -16 \\ & & -20 & -4 & 16 \\ \hline & 5 & 1 & -4 & 0 \end{array}$$

(b) Write your answer in the following form: Quotient + $\frac{\text{Remainder}}{x + 4}$.

$$\frac{5x^3 + 21x^2 - 16}{x + 4} = 5x^2 + x - 4 + \frac{0}{x + 4}$$

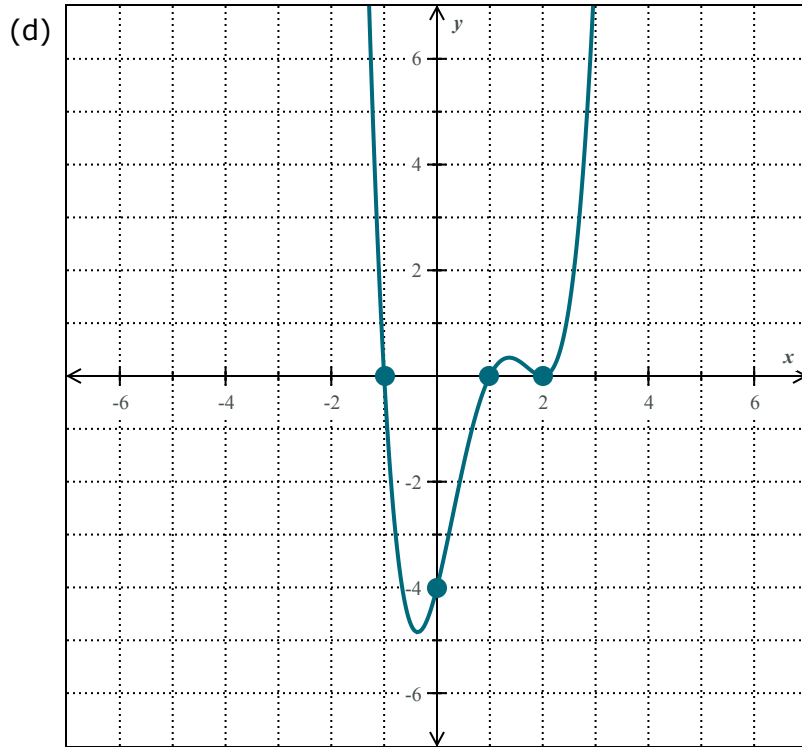
Question 18 of 40

(a) ▼

(b) Zero(s) where the graph crosses the x -axis:

Zero(s) where the graph touches, but does not cross the x -axis:

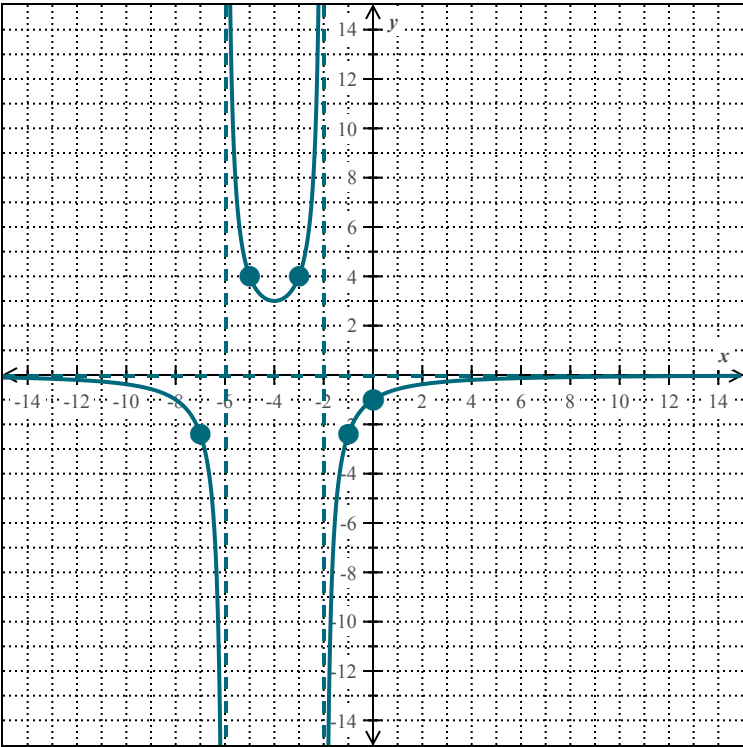
(c) y -intercept:



Question 19 of 40

$$h(x) = (x + 2)(x - 1)(x - 3)$$

Question 20 of 40



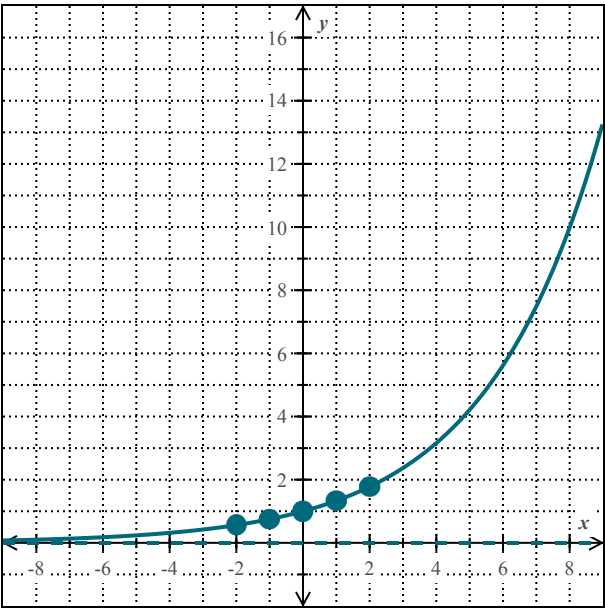
Question 21 of 40

$(-7, 4]$

Question 22 of 40

$(-\infty, -4] \cup [1, 5]$

Question 23 of 40



Question 24 of 40

Initial amount: 2,400 grams

Amount after 40 hours: 331 grams

Question 25 of 40

(a) $7^{-1} = \frac{1}{7}$

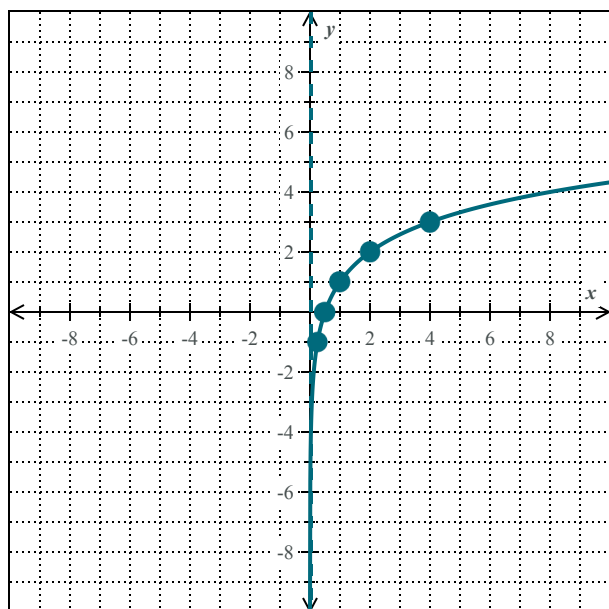
(b) $\log_2 32 = 5$

Question 26 of 40

(a) $\log_9 1 = 0$

(b) $\log_8 \frac{1}{64} = -2$

Question 27 of 40



Question 28 of 40

$$\log_4 7 - \log_4 11 = \log_4 \frac{7}{11}$$

$$\log_5 11 + \log_5 9 = \log_5 99$$

$$\log_2 \frac{1}{25} = -2\log_2 5$$

Question 29 of 40

$$\log \left(x^5 \sqrt[3]{y^2 z} \right) = 5 \log x + \frac{2}{3} \log y + \frac{1}{3} \log z$$

Question 30 of 40

$$\log_5 \left((2z + 1)^7 (z + 6)^{\frac{1}{4}} \right)$$

Question 31 of 40

11.7 hours

Question 32 of 40

$$x = 16.09$$

Question 33 of 40

$$x = -3$$

Question 34 of 40

$$\begin{array}{l} x = 3 \\ y = -5 \end{array}$$

Question 35 of 40

Length of each Plan A workout: 1.25 hour(s)

Length of each Plan B workout: 1.25 hour(s)

Question 36 of 40

(a) For each step below, enter the coefficient for the row operation. The first matrix in Step 1 is the augmented matrix for the given system of equations.

Step 1:

$$\left[\begin{array}{cc|c} 2 & -12 & 16 \\ 4 & -19 & 22 \end{array} \right] \xrightarrow{\frac{1}{2} \cdot R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & -6 & 8 \\ 4 & -19 & 22 \end{array} \right]$$

Step 2:

$$\left[\begin{array}{cc|c} 1 & -6 & 8 \\ 4 & -19 & 22 \end{array} \right] \xrightarrow{-4 \cdot R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 5 & -10 \end{array} \right]$$

Step 3:

$$\left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 5 & -10 \end{array} \right] \xrightarrow{\frac{1}{5} \cdot R_2 \rightarrow R_2} \left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 1 & -2 \end{array} \right]$$

Step 4: Enter the coefficient for the row operations and the missing entries in the resulting matrix.

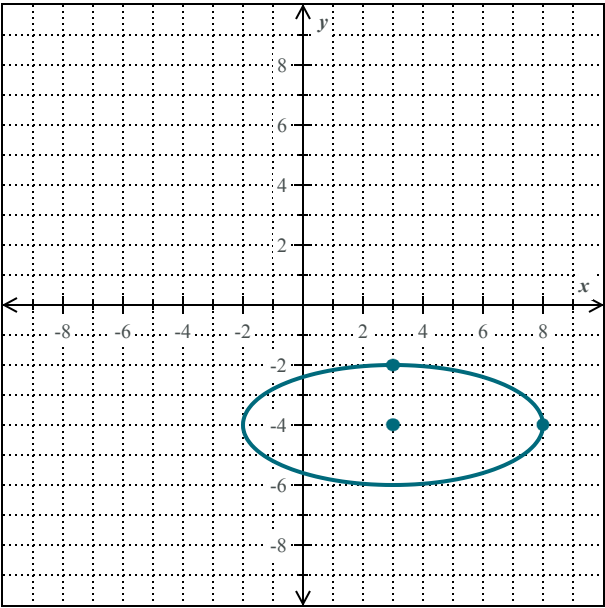
$$\left[\begin{array}{cc|c} 1 & -6 & 8 \\ 0 & 1 & -2 \end{array} \right] \xrightarrow{6 \cdot R_2 + R_1 \rightarrow R_1} \left[\begin{array}{cc|c} 1 & 0 & -4 \\ 0 & 1 & -2 \end{array} \right]$$

(b) Give the solution.

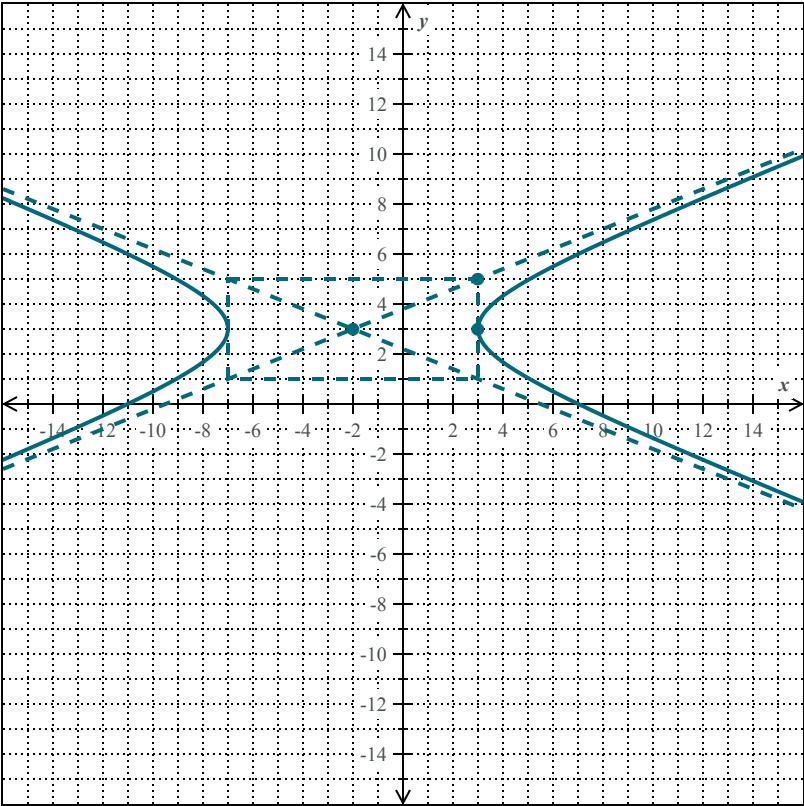
$$x = -4$$

$$y = -2$$

Question 37 of 40



Question 38 of 40



Question 39 of 40

$$\frac{8}{3} \text{ ' } \frac{14}{9} \text{ ' } \frac{32}{27} \text{ ' } \frac{86}{81}$$

Question 40 of 40

$$\left(\frac{1}{3}\right) + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^3 + \dots + \left(\frac{1}{3}\right)^7 = \frac{1093}{2187}$$

$$\sum_{j=1}^8 2(-4)^j = 104,856$$